Mechanics Year 2

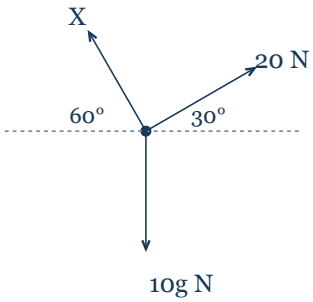
Chapter 7: Applications of Forces

Overview 2

Combine friction, resolving forces, $F = ma$, inclined planes and connected particles. These preview examples are developed in the chapter.

1. Equilibrium

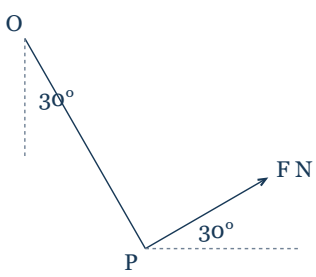
Resolve forces to determine unknown magnitudes.



The supplied numerical sketch is inconsistent with equilibrium for $g = 9.8$; its arrows illustrate resolving forces.

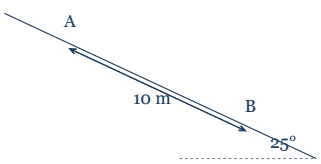
2. Weight and tension

A $2\,\text{kg}$ particle on a light string is held at 30° to the downward vertical by a force F, 30° above horizontal. Find F and the tension. (8 marks)



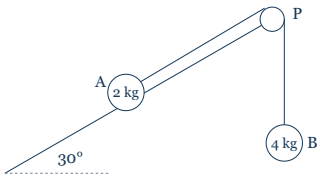
3. Inclined motion

A $0.6\,\text{kg}$ particle slides $10\,\text{m}$ down a rough 25° plane in $3.5\,\text{s}$, starting at $2\,\text{m s}^{-1}$. Find final speed, acceleration and friction coefficient. (3, 2, 5 marks)



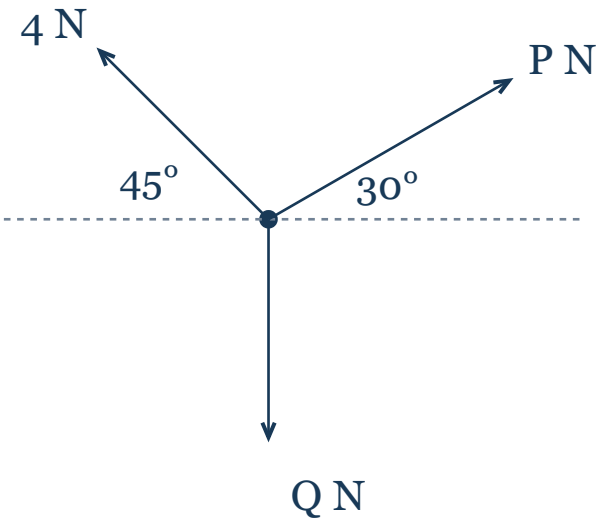
4. Connected particles

A $2\,\text{kg}$ particle on a rough 30° slope is connected by a taut light inextensible string over a smooth fixed pulley to a hanging $4\,\text{kg}$ particle. With $\mu = 1/\sqrt{3}$, release from rest makes A move uphill. Find tension immediately after release. (9 marks)



Finding Unknown Forces by Resolving 3

The particle is in equilibrium under the forces shown. Resolve horizontally and vertically to find the magnitudes P and Q.

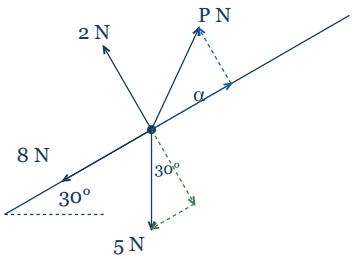


Resolve separately in the two directions, as before.

Working

Unknown Forces on Inclined Planes 4

The particle is in equilibrium on the inclined plane under the forces shown. Find P and α.



Use dotted, labelled arrows for resolved components. Blue arrows resolve P; green arrows resolve the weight.

Working

Resolve the applied force:

P parallel

P normal

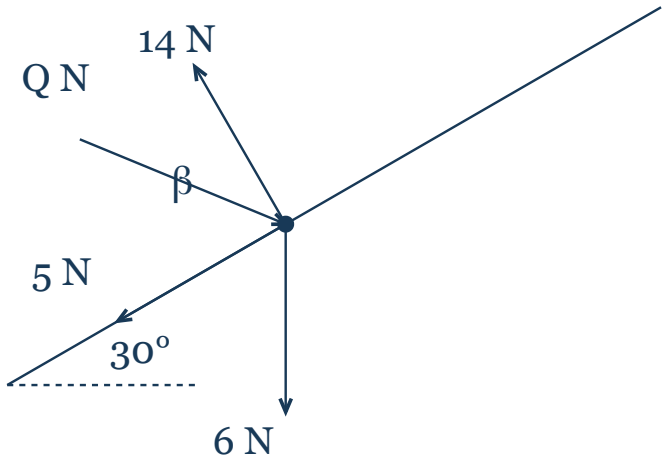
Resolve the weight:

Weight normal

Weight parallel

Test Your Understanding 5

The particle is in equilibrium under the forces shown. Find Q and β.



Resolve

Solve

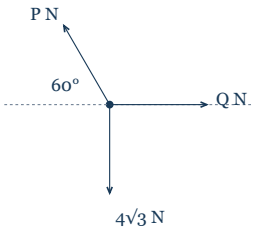
Exercise 7A — Y2 Mechanics Pages 130–132 6

2. For each particle in static equilibrium:

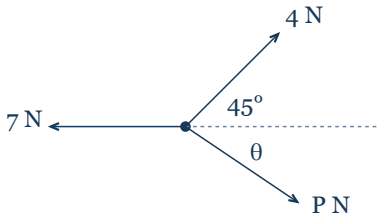
i. Draw a triangle of forces.

ii. Use trigonometry to find unknown forces P, Q and unknown angle θ.

a.



b.



a.i

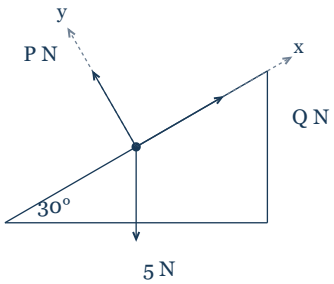
b.i

a.ii

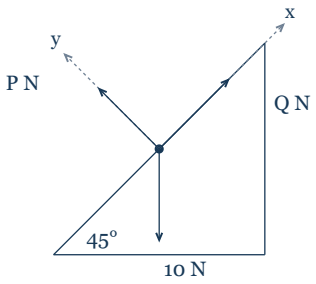
b.ii

3. Each particle rests in equilibrium on the plane under the forces shown. Find P and Q.

a.



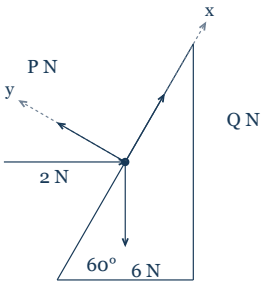
b.



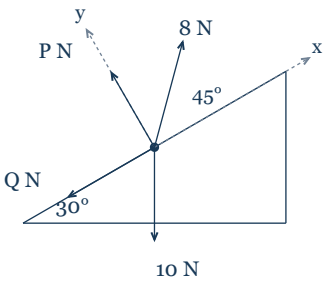
a

b

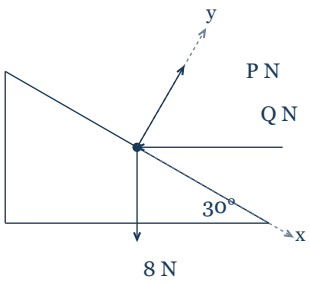
3c.



3d.



3e.



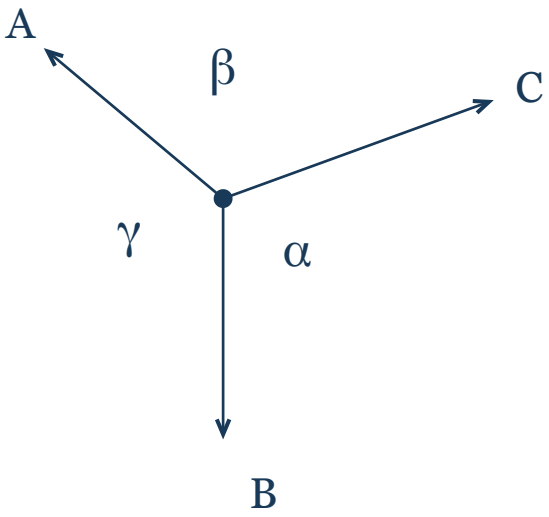
c

d

e

Challenge

Three coplanar forces of magnitudes A, B, C act on a particle in equilibrium, as shown. Prove $\dfrac{A}{\sin\alpha} = \dfrac{B}{\sin\beta} = \dfrac{C}{\sin\gamma}$.



Proof

Proof

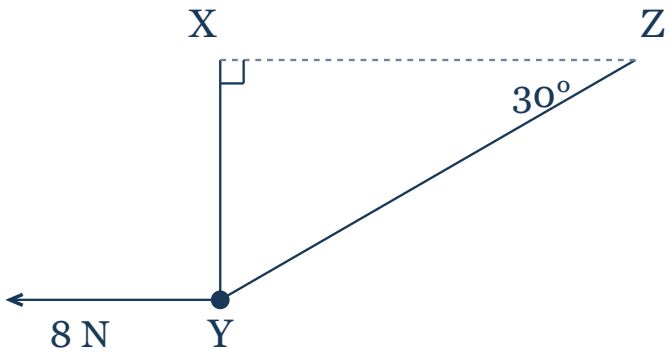
Proof

Proof

Proof

Modelling with Statics 10

Apply resolving to tension, weight and pulleys. A smooth bead Y is threaded on a light inextensible string whose ends are fixed at X, Z, at the same horizontal level. A horizontal 8 N force parallel to ZX holds the bead in equilibrium, with Y vertically below X and $\angle XZY = 30°$. Find the tension and the bead's weight.



Draw the force diagram.

Diagram

The smooth bead allows the tension to be the same in both parts of the light string.

Tension

Weight

Further Example 11

A $3\,\text{kg}$ mass rests on a smooth plane at 45° to horizontal. A cable runs up the line of greatest slope over a smooth pulley and supports a freely hanging $1\,\text{kg}$ mass. Model the masses as particles and the cable as a light inextensible string. A horizontal force P acts towards the plane on the $3\,\text{kg}$ mass; the system is in equilibrium. Find: (a) P; (b) the normal reaction; (c) how the smooth-pulley assumption is used.

Draw the force diagram.

Diagram

a.i

b

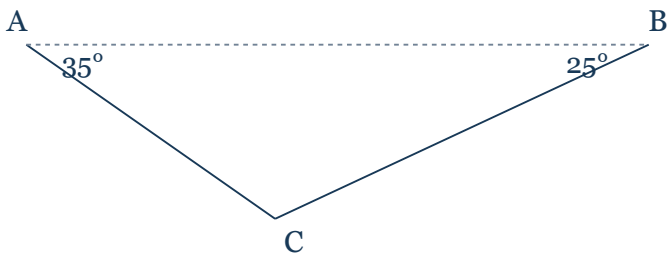
a.ii

c

Test Your Understanding 12

Edexcel M1 (Old), May 2013(R), Q2

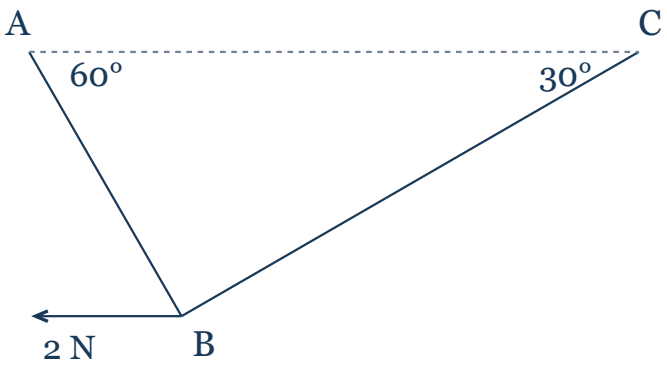
A particle of weight $8\,\text{N}$ is attached at C to two light inextensible strings AC, BC. Their other ends are fixed to a horizontal ceiling at A, B. The particle rests in equilibrium; the strings lie in a vertical plane, at 35° and 25° to horizontal respectively. Find (i) the tension in AC; (ii) the tension in BC.



The particle cannot slide along the string: these are two separate strings with separate tensions. Introduce T_A, T_B.

Working

5. A smooth bead B is threaded on a light inextensible string fixed at A, C, at the same horizontal level. A horizontal $2\,\text{N}$ force parallel to CA holds it in equilibrium. The string sections make angles $60°, 30°$ with horizontal. Find (a) the tension; (b) the bead's mass; (c) state how the smooth-bead assumption is used.



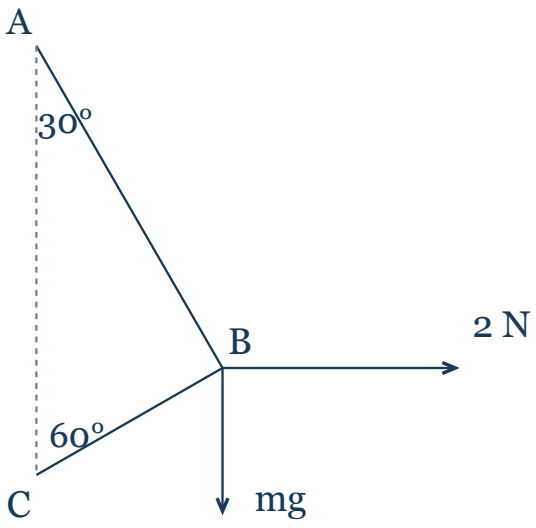
a

a

b

c

6. A smooth bead B is threaded on a light inextensible string fixed at A, C, where A is vertically above C. A horizontal $2\,\text{N}$ force holds the bead in equilibrium. Sections AB, BC make angles $30°, 60°$ with vertical respectively. Find (a) the tension; (b) the bead's mass, to the nearest gram.

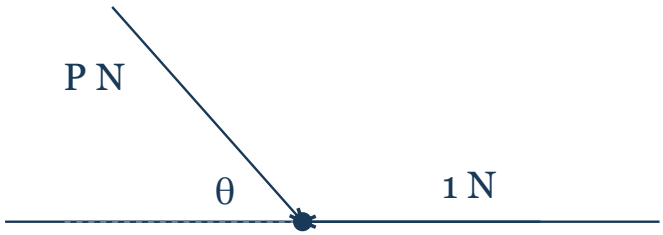


a

a

b

7. A particle of weight $2\,\text{N}$ rests on a smooth horizontal surface in equilibrium under the two external forces shown. One is a horizontal $1\,\text{N}$ force; the other has magnitude P, at angle θ to horizontal, where $\tan\theta = 12/5$. Find (a) P; (b) the normal reaction.

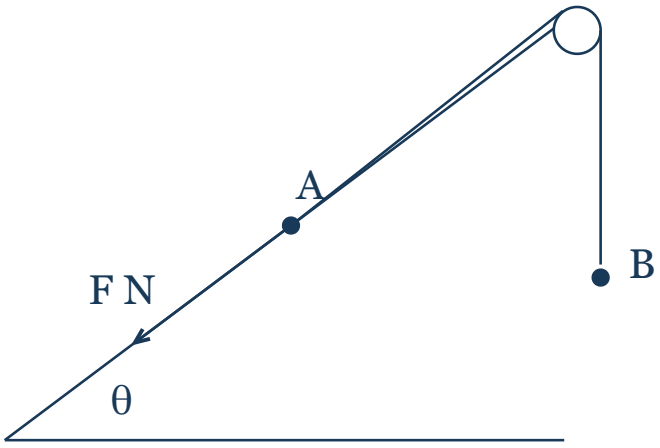


a

a

b

11. Particle A, mass $2\,\text{kg}$, rests in equilibrium on a smooth plane at angle θ to horizontal, where $\tan\theta = 3/4$. A light inextensible string runs from A over a smooth pulley to hanging particle B, mass $5\,\text{kg}$. A force F acts on A down the line of greatest slope. Find (a) the normal reaction; (b) F; (c) state how the smooth-pulley assumption is used.



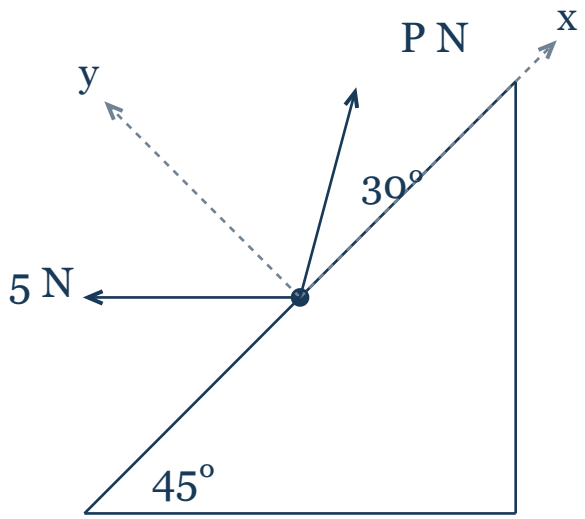
a

a

b

c

12. A particle of weight $20\,\text{N}$ rests in equilibrium on a smooth 45° inclined plane. A horizontal $5\,\text{N}$ force acts to the left; force P acts at 30° above the plane. Find the normal reaction.



Working

Working

Friction 18

Friction satisfies $F \le \mu R$, with $F = \mu R$ in limiting equilibrium or sliding under this model. At rest, friction opposes the motion that would occur without it.

A $10\,\text{kg}$ box rests in limiting equilibrium on a rough plane at 20° above horizontal. (a) Find μ. A horizontal force P is then applied towards the plane. (b) Find the maximum P for equilibrium.

a diagram

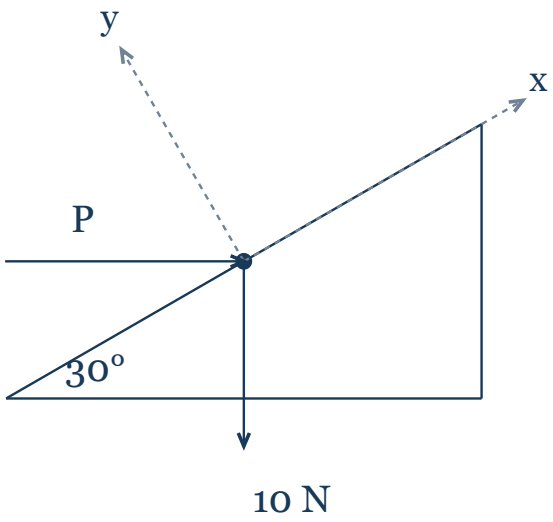
b diagram

a working

b working

Edexcel M1 (Old), January 2006, Q5

A parcel of weight 10 N lies on a rough 30° inclined plane. Horizontal force P acts towards the plane. It is in equilibrium on the point of slipping uphill; the normal reaction is 18 N, coefficient of friction μ. Find (a) P (4 marks); (b) μ (5 marks). The horizontal force is removed. (c) Determine whether the parcel moves (5 marks).



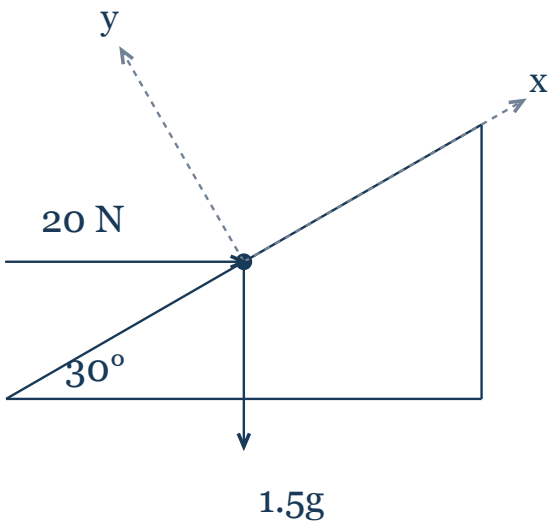
a

a

b

c

8. A horizontal 20 N force acts on a 1.5 kg block resting in equilibrium on a rough 30° plane. The force acts in the vertical plane containing the line of greatest slope. Find (a) the normal reaction; (b) the friction magnitude and direction; (c) the minimum coefficient of friction.

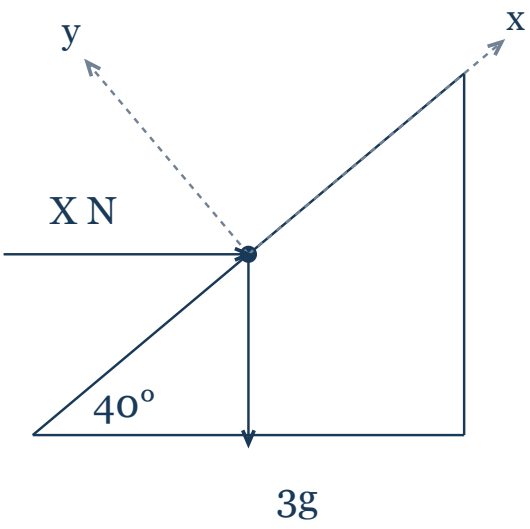


Working

Working

Working

9. A $3\,\text{kg}$ box lies on a rough plane inclined at 40° to horizontal. A horizontal force X holds it in equilibrium, acting in the vertical plane containing the line of greatest slope. The coefficient of friction is 0.3; the box is in limiting equilibrium about to move uphill. Find (a) X; (b) the normal reaction.



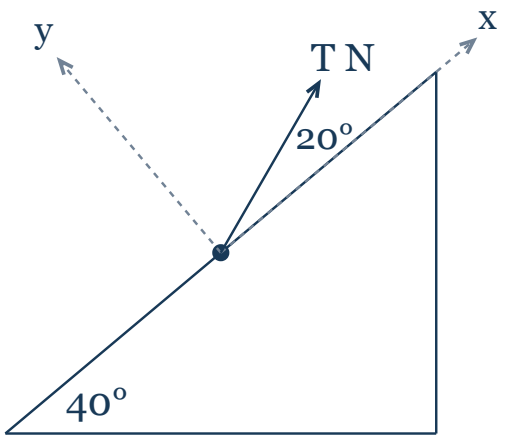
a

a

a

b

11. A $0.5\,\text{kg}$ box lies on a 40° inclined plane with coefficient of friction $1/5$. A light string keeps it in equilibrium, lying in the vertical plane containing the line of greatest slope, at 20° above the plane. Model the box as a particle. The tension is T. Find its possible range, with limiting equilibrium at the endpoints.



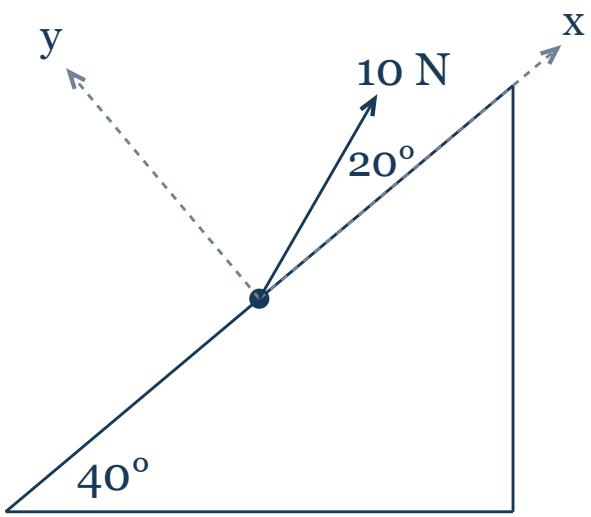
Working

Working

Working

Working

12. A 1 kg box lies on a plane at 40° to horizontal. A light string, at 20° above the plane in the vertical plane containing its line of greatest slope, holds the box in equilibrium on the point of moving uphill. Model it as a particle. The tension is 10 N. Find the coefficient of friction μ.



Working

Working

13. A 2 kg particle rests in limiting equilibrium on a rough plane at angle θ above horizontal, where $\tan\theta = 3/4$. A horizontal force P, directed into the plane, is then applied. Given that the particle remains in equilibrium, find the maximum possible P.

Working

Working

Working

Static Rigid Bodies 25

For a stationary rigid body:

• The resultant force is zero.

• The resultant moment is zero.

These are the moment problems already studied, now including friction.

A uniform rod AB, mass $40\,\text{kg}$, length $10\,\text{m}$, has end A on rough horizontal ground and rests against a smooth peg C, where $AC = 8\,\text{m}$. It is in limiting equilibrium at 15° to horizontal. Find (a) the reaction at C; (b) the ground's coefficient of friction.

Diagram

Diagram

a. Take moments about A.

a

b. Resolve horizontally, then vertically.

b.i

b.ii

Further Example 26

A ladder AB, mass m, length $3a$, rests with A on rough horizontal ground and B against a smooth vertical wall. A load of mass $2m$ is fixed at C, where $AC = a$. Model the ladder as a uniform rod in a vertical plane perpendicular to the wall, and the load as a particle. The ladder rests in limiting equilibrium at 60° to the ground. (a) Find the coefficient of friction with the ground. (b) State how the uniform-ladder assumption is used.

Diagram

Two different reaction forces need different variables.

a.i

a.ii

a.iii

b

4. A uniform ladder, mass $20\,\text{kg}$, length $8\,\text{m}$, rests against a smooth vertical wall with its lower end on rough horizontal ground. The ground's coefficient of friction is 0.3; the ladder makes angle θ with horizontal, where $\tan\theta = 2$. A $30\,\text{kg}$ boy climbs it. Model the ladder as a uniform rod and the boy as a particle. (a) How far can he climb before slipping occurs? (b) Criticise the model for (i) the ladder, (ii) the wall.

a

a

b.i

b.ii

8. A uniform ladder rests in limiting equilibrium with one end on rough horizontal ground and the other against a rough vertical wall. The coefficients of friction are μ_1 at the ground and μ_2 at the wall. It makes angle θ with horizontal. Show

$$\tan\theta = \frac{1 - \mu_1\mu_2}{2\mu_1}.$$

Working

Working

Working

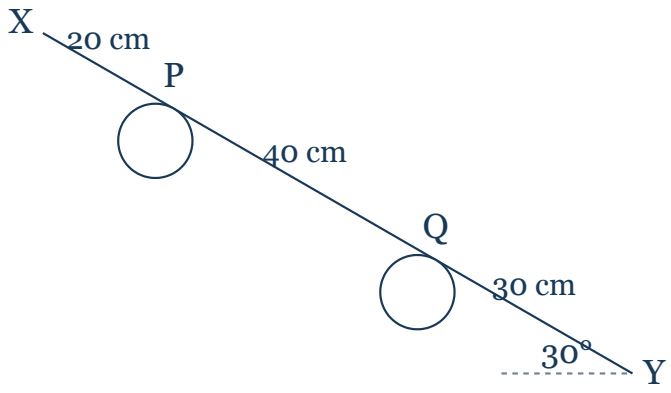
9. A uniform ladder of weight W rests in equilibrium on rough horizontal ground against a smooth vertical wall. Its vertical plane is perpendicular to the wall; the ladder makes 60° with horizontal. The ground's coefficient of friction is μ. (a) Find the wall force in terms of W. (b) Show that $\mu \geq \sqrt{3}/6$.

a

a

b

10. A uniform rod XY, weight $20\,\text{N}$, length $90\,\text{cm}$, rests on two parallel pegs with X above Y, in a vertical plane perpendicular to the peg axes. It makes 30° with horizontal and touches the pegs at P, Q, where $XP = 20\,\text{cm}$, $XQ = 60\,\text{cm}$. (a) Find the normal force components at P, Q. The friction coefficient at each peg is μ. (b) Given impending slip, find μ.



a

a

b

Particles Moving on a Rough Plane 31

We have considered motion on smooth planes and equilibrium on rough planes. Now consider motion on rough planes.

A particle is held at rest on a rough plane inclined at α to horizontal, where $\tan\alpha = 3/4$. The coefficient of friction is 0.5. Released, it slides downhill. Find (a) its acceleration; (b) the distance travelled in the first $2\,\text{s}$.

Diagram

a

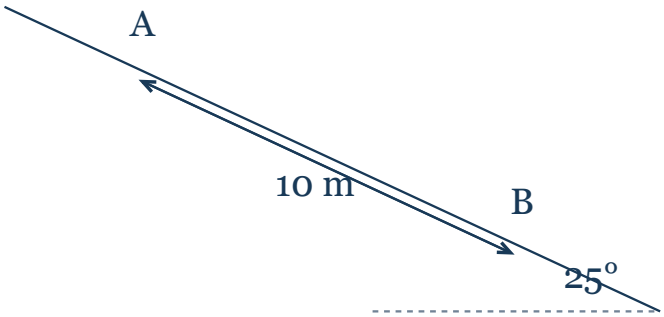
a

b

Test Your Understanding 32

Edexcel M1 (Old), May 2013(R), Q5

Particle P, mass $0.6\,\text{kg}$, slides with constant acceleration down the line of greatest slope of a rough plane at 25° to horizontal. It passes A, B, where $AB = 10\,\text{m}$, taking $3.5\,\text{s}$ from A to B. Its speed at A is $2\,\text{m}\,\text{s}^{-1}$. Find (a) its speed at B; (b) its acceleration; (c) the coefficient of friction.



a

b

c

7. A $10\,\text{kg}$ parcel is released from rest on a rough plane inclined at 25° to horizontal. (a) Find the normal reaction. After $2\,\text{s}$, it has moved $4\,\text{m}$ downhill. (b) Find the coefficient of friction.

a

a

b

8. Particle P is projected up a rough plane at angle α to horizontal, where $\tan\alpha = 3/4$. The friction coefficient is $1/3$. From A, initial speed is $20\,\text{m}\,\text{s}^{-1}$; it stops momentarily at B. (a) Show its uphill deceleration is $13g/15$. Find (b) AB, (c) the time from A to B, both to 2 s.f.; (d) its speed on returning to A.

a

b

d

a

c

d

9. A $2\,\text{kg}$ particle is released from rest on a rough slope at angle α to horizontal, where $\tan\alpha = 2/5$. After $3\,\text{s}$, its speed is $6\,\text{m s}^{-1}$. Find the coefficient of friction μ.

Working

Working

Working

11. A $5\,\text{kg}$ particle is projected up a rough 10° slope at $16\,\text{m s}^{-1}$, stopping at P after $5\,\text{s}$. (a) Find μ. (b) State, with supporting calculations, whether it remains at rest or slides back downhill.

a

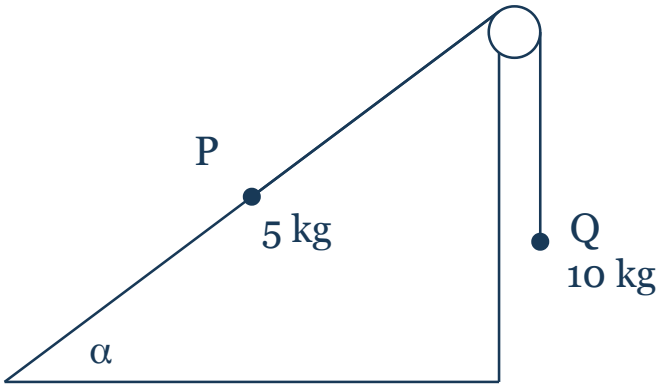
a

b

Connected Particles Involving Friction 37

Connected-particle problems from Mechanics Year 1 now include friction.

Particles P, Q, masses $5\,\text{kg}, 10\,\text{kg}$, are connected by a light inextensible string over a small smooth fixed pulley at the top of a rough plane. P rests on the plane; Q hangs freely with the string vertical and taut. The slope angle is α, with $\tan\alpha = 0.75$, and friction coefficient for P is 0.2. Released from rest, find (a) the acceleration; (b) the tension.



Diagram

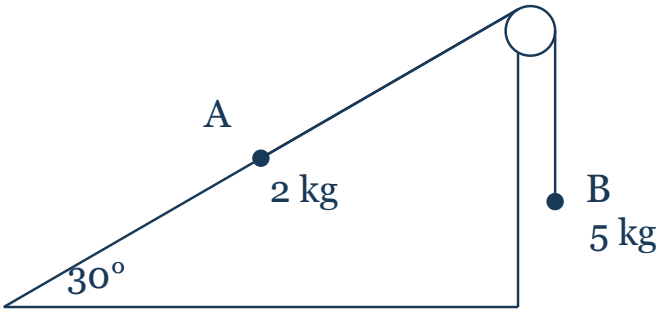
a

a

b

Further Example 38

A light inextensible string joins block A, mass $2\,\text{kg}$, on a smooth fixed 30° plane to block B, mass $5\,\text{kg}$, hanging freely. It lies along the greatest slope and passes over a smooth light pulley fixed at the top. Release from rest. Model the blocks as particles and ignore air resistance. (a)(i) Show B's acceleration is $4g/7$; (ii) find tension. (b) How is the inextensible-string assumption used? (c) Find the magnitude of the string's force on the pulley.



Diagram

a

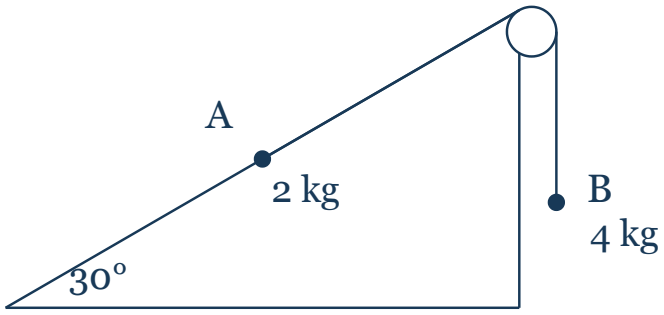
c

b

c

Edexcel M1 (Old), May 2013(R), Q3

A rough fixed plane is inclined at 30° to horizontal. A small smooth pulley P is fixed at its top. Particles A, B, masses 2 kg, 4 kg, are joined by a light inextensible string over P. The string from A to P is parallel to the greatest slope; B hangs freely. The friction coefficient between A and the plane is $1/\sqrt{3}$. Initially held at rest, release them with string taut and A moving uphill. Find the tension immediately after release. (9 marks)



Working

Working

3. Particles P, Q, masses 2 kg, 3 kg, are connected by a light inextensible string over a small smooth fixed pulley at the top of a rough 30° plane. P is held at rest on the plane; Q hangs freely with string vertical and taut. On release, P accelerates uphill at $2.5\,\text{m}\,\text{s}^{-2}$. Find (a) tension; (b) the coefficient of friction; (c) the string's force on the pulley.

a

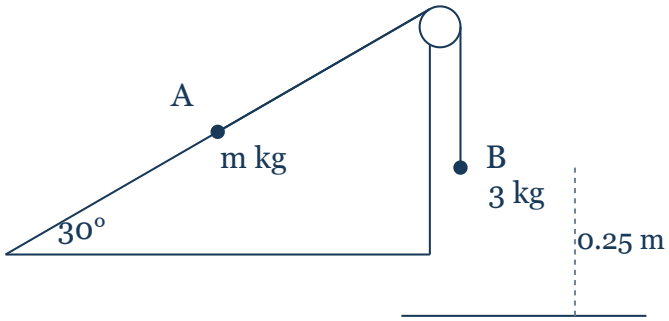
b

a

c

c

4. Particles A, B, masses m kg, 3 kg, are connected by a light inextensible string over a smooth fixed pulley P. A is held on a smooth fixed 30° plane, with string AP along its greatest slope; B hangs freely. Release from rest with B 0.25 m above horizontal ground. B initially accelerates downwards at $2g/5$. A does not reach P. Find (a) tension while B descends; (b) m. B hits the ground without rebounding. (c) Find the time from impact until A reaches its highest point.



a

b

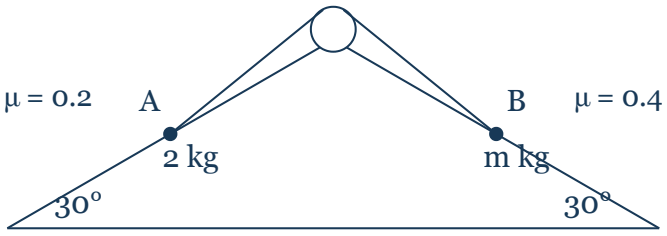
c

a

c

5. Particles A, B on back-to-back rough 30° slopes are connected by a light inextensible string over a smooth pulley. A has mass 2 kg; B has mass m kg. The friction coefficients are 0.2 for A, 0.4 for B. Show the maximum m before motion begins is

$$\frac{10+2\sqrt{3}}{5-2\sqrt{3}}.$$

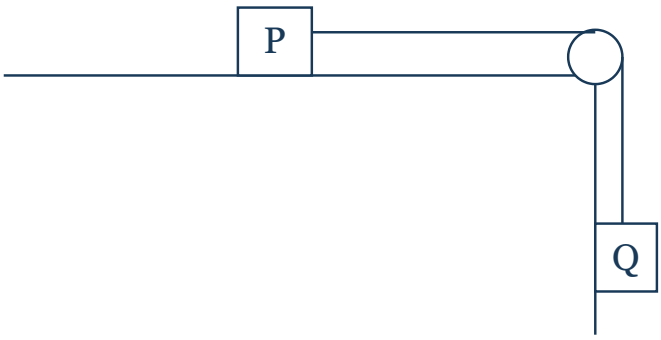


Working

Working

Working

6. A metal block P, mass $1.5\,\text{kg}$, rests on a rough horizontal workbench. A light inextensible string joins it over a small smooth pulley fixed at the edge to freely hanging box Q, mass $1.6\,\text{kg}$. The coefficient of friction for P is μ. Release from rest with the string taut; after $2\,\text{s}$, Q's speed is $6\,\text{m s}^{-1}$. Model both as particles. (a) Find acceleration of Q; (b) tension; (c) show $\mu = 0.434$ (3 s.f.); (d) how is inextensibility used?



a

b

b

c

d